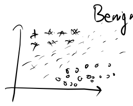
Support Vector Machine (SVM)

Benign

Malignant

F1 → tumour radius
F2 → tumour texture

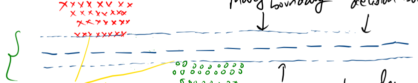
Benign Malignant



Class 0

margin boundary

decision boundary

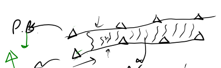


margin boundary.

Class 1

Support vectors

maximum margin boundary.

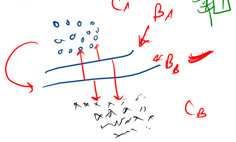


P.B

JB

No man's lands.

Maximum margin principle



C_A

B_A

B_B

C_B

12 →

29th of July.

fasting

29th of Aug.

A

B

↓ interpret.

Tiger, Cheetah, Leopard, Jaguar, Panther.

−1

+1 observation

hard margin svm.

Real world data is rarely this clean.

Soft margin svm

C → strictness about training errors.

Large C → Training mistakes are very expensive.

small C → willing to tolerate some training errors.

Large C → less tolerance for violations → narrow margin
↓ more complex model.

small C → more ___ ___ ___ → wider margin
↓ strong regularization.

Gmail → Spam / Not spam

Congratulations! You won an award. Click here for details.

→ Genuine email
→ spam email

High

→ low C, svm more flexible separation.

hinge loss

vs.

+1 −1

loss = 0 → correct & outside margin

loss → positive loss → inside margin & correct.

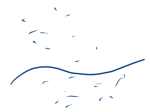
loss → large positive → incorrectly classified.

x & y, x^2, y^2, xy ___

Kernel trick.

2D

$x \to \phi(x)$.



computation → expensive.

→ $k(x,z) = \phi(x).\phi(z)$

RBF Kernel Intuition

create flexible local decision boundary.

Svm. → 20 minutes.

γ → Gamma.

small gamma • larger region. smooth.

large gamma • small region. complex boundaries.

Age.

Small gamma → broad influence → smoother boundary.

large gamma → narrow influence → complex boundary.

$y = mx + c$ → slope intercept form.

bias ↓

$w.x + b = 0$

↓

$w_1x_1 + w_2x_2 + b = 0$

$f(x) = w.x + b$

weight vector.

$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$w_1x_1 + w_2x_2 + b = 0$

$w_1x + w_2y + b = 0$

$\min \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$

L. C → strong penalty for violation

S. C → greater tolerance for violation.

$x \to \phi(x)$

SVC(kernel = "linear")   linear boundary.

$ax^2 + bx + c = 0$

SVC(kernel = "poly", degree = 2)
↓
5

flexibility increase, overfitting risk.

C & Gamma.

↓ Punish training violation

↓ influence.

Small C + Small γ → strong regularization & smooth boundary.

Large C + Small γ → less tol. for training error but smooth geometry.

Small C + large γ → model allow some error but local influence.

Large C + large γ → model tried to fit training data & local influence.
↳ severe overfitting.

SVM for regression – SVR

Support vector Regression:–

epsilon :–

upper epsilon boundary
predictions
lower epsilon boundary

epsilon insensitive loss.

☆ – if the prediction error is small then epsilon loss is zero.

SVR(kernel="rbf", c=1.0, epsilon=0.1)

30 inch waist   actual → 30.5

38 inch.

acceptable room of small mistakes.

100000

100001

SVR.

margin of tolerance.

H.

office

9:30

10

30 minutes.

$(\epsilon) = 5$ minutes.

25 & 35

9:45 10:15

D1 → 32 minutes.

D2 → 28 minutes.

D3 → 50 minutes.

real error.

model.predict(X)

svc(probability = True)